

Protecting groups are utilized in organic synthesis to mask a particular functional group so that it does not react during a subsequent reaction. The protecting group should be easily added and selectively removed to avoid any undesired side reactions. There are many protecting group options for each functional group.

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Installation of an acetonide protecting group to a diol using acetone and toluenesulfonic acid (TsOH) (an acid catalyst) is shown in Figure 1.

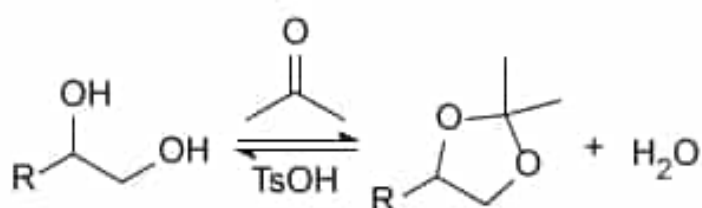


Figure 1 General acetonide protection of a diol (Note: R = any alkyl group)

Acetonides are particularly useful in carbohydrate synthesis. In one instance, researchers used an acetonide protecting group on a mannose derivative (Figure 2) and a galactose derivative (Figure 3). Acetonides can be removed using acid via a hydrolysis reaction.

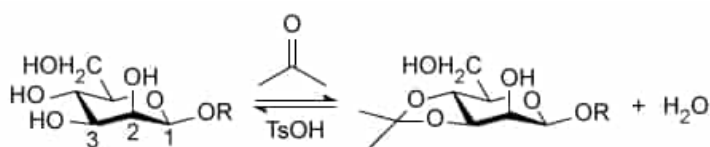


Figure 2 Acetonide protection of a mannose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

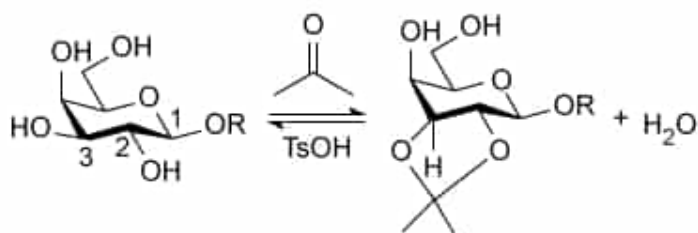


Figure 3 Acetonide protection of a galactose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

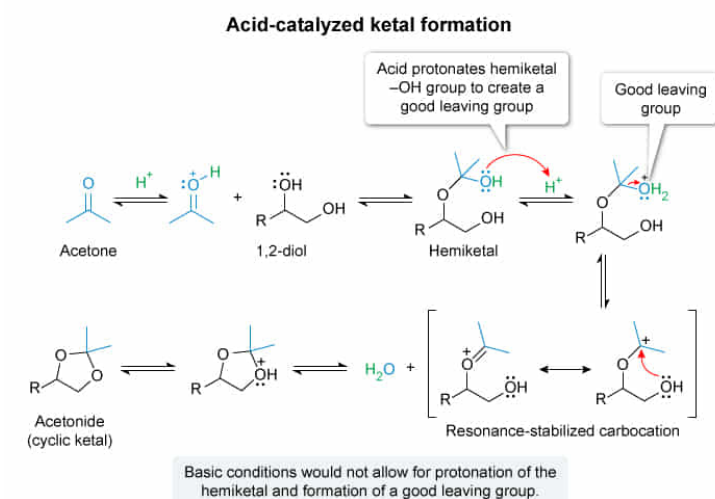
Which of the following best explains why a catalytic base CANNOT be used in the formation of an acetonide as shown in Figure 1?

- ☐ A. An alkoxide cannot add to the carbonyl to create a hemiketal under basic conditions. [18%]
- ☐ B. An alkoxide cannot add to the carbonyl to create a hemiacetal under basic conditions. [13%]
- ✓ ☒ C. The hemiketal –OH group cannot be protonated under basic conditions to make a good leaving group for the ketal to form. [42%]
- ☐ D. The hemiacetal –OH group cannot be protonated under basic conditions to make a good leaving group for the acetal to form. [26%]

Correct

41% answered correctly

Explanation:



Ketals, which are often used as protecting groups, are formed by acid-catalyzed **nucleophilic addition** of two alcohols to a ketone carbonyl (C=O). The first step is formation of the **hemiketal**. Acid protonates the

carbonyl oxygen, making the carbonyl carbon more electrophilic and susceptible to nucleophilic attack by the alcohol. The hemiketal forms upon deprotonation of the newly added alcohol.

While hemiketal formation can be **acid- or base-catalyzed**, the second step (formation of a ketal) **must be acid-catalyzed**. Acid protonates the hydroxyl oxygen to form H_2O , a **good leaving group**. Dehydration (elimination of water) occurs to generate a resonance-stabilized carbocation. A second alcohol adds to the carbocation and is deprotonated to form a **ketal**. **Hemiacetals** and **acetals** are formed through an analogous process using an aldehyde rather than a ketone. Cyclic ketals and cyclic acetals are formed when the alcohol is a **1,2- or 1,3-diol**, following a similar mechanism.

The passage states that acetonides are cyclic protecting groups for 1,2- or 1,3-diols. Because Figure 1 shows the ketone acetone is used to form an acetonide, this protecting group is a **cyclic ketal**. Formation of an acetonide must be acid-catalyzed because the hemiketal hydroxyl must be protonated to create a good leaving group prior to the nucleophilic addition of the second -OH group and formation of the ketal. Therefore, there are **no acidic protons** to protonate -OH to **create a good leaving group** for the ketal to form **under basic conditions**.

(Choices A and B) Formation of hemiketals and hemiacetals *can* occur under basic conditions because alkoxides can add to a carbonyl. The reaction would not be able to proceed to the next step under basic conditions.

(Choice D) Because the acetonide is formed using a ketone (acetone), it is a *ketal* rather than an acetal.

Educational objective:

Ketals and acetals are formed by acid-catalyzed nucleophilic addition of two alcohols to a ketone or aldehyde carbonyl, respectively. An acid catalyst must be used to protonate the hemiketal (or hemiacetal) -OH group to create a good leaving group so the second alcohol molecule

can be added.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403990

System:
Functional Groups and Their Reactions

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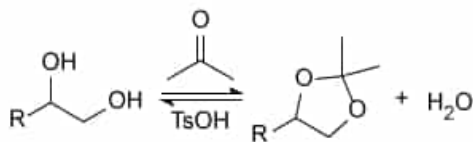


Figure 1 General acetonide protection of a diol (Note: R = any alkyl group)

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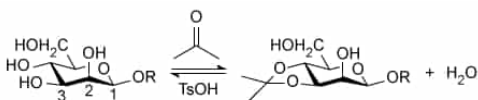


Figure 2 Acetonide protection of a mannose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

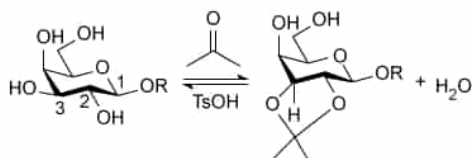
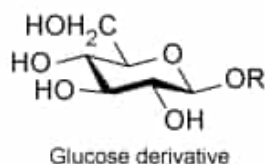


Figure 3 Acetonide protection of a galactose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

A glucose derivative is shown below.

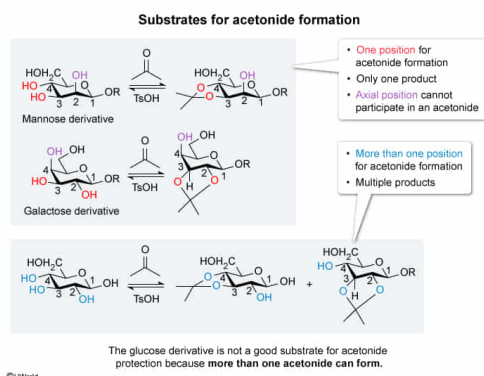


If researchers desire only one product when adding an acetonide protecting group to a monosaccharide derivative as shown in figures 2 and 3, the glucose derivative is a poor reaction substrate because:

- ☒ A. multiple acetonides can form. [68%]
- ☐ B. only one acetonide can form. [3%]
- ☐ C. the reaction is regioselective. [10%]
- ☐ D. the reaction is stereoselective. [17%]

Correct
67% answered correctly

Explanation:



Many organic molecules contain multiple **functional groups** that react under the same reaction conditions. If a reaction is desired at only one position or functional group, a **protecting group** can be installed at the positions where no reaction is desired. Protecting groups allow for **selective reaction** at a desired position in a molecule and prevent undesired reactions and products. A protecting group should be stable to the conditions of the desired transformation before being removed in a subsequent reaction.

Hydroxyl (–OH) groups are often protected as **silyl ethers**, and **diols** can be protected as acetonides as described in the passage. Figures 2 and 3 show the addition of an acetonide protecting group to mannose and galactose derivatives, respectively. The mannose and galactose derivatives are depicted in chair conformations. Examining the spatial position (**axial or equatorial**) of the –OH groups on the six-membered ring:

- **Mannose derivative:** Carbon 2 (axial –OH) and carbons 3 and 4 (equatorial –OH)
- **Galactose derivative:** Carbons 2 and 3 (equatorial –OH) and carbon 4 (axial –OH)

The acetonide protecting group can form when adjacent –OH groups have an accommodating geometric relationship. In carbohydrates, acetonides can *only* form between hydroxyl groups that are in an equatorial position—axial hydroxyl groups do not have the correct geometry to form the cyclic structure. The –OR group on carbon 1 cannot react to form an acetonide because it is not a hydroxyl group. The question asks why a glucose derivative would not be a good substrate for acetonide protection if only one product is desired. The given glucose derivative has carbons 2, 3, and 4 with equatorial –OH groups. As a result, acetonides could form between –OH groups on carbons 2 and 3 OR carbons 3 and 4.

Therefore, the glucose derivative is a poor substrate for acetonide protection because **more than one acetonide can form**.

(Choice B) Forming only one acetonide *would* make the glucose derivative a good substrate for the reaction.

(Choices C and D) Because more than one acetonide can form, the reaction is *neither regioselective nor stereoselective*.

Educational objective:

Some organic molecules contain multiple functional groups that react under the same conditions. For a reaction to selectively occur at a desired position in a molecule, protecting groups are added and then later removed so certain functional groups do not react.

Time spent: 0 sec
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Organic Chemistry

QID: 403991
System:
Functional Groups and Their Reactions

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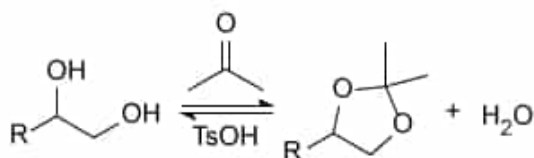


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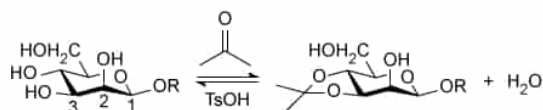


Figure 2 Acetonide protection of a mannose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

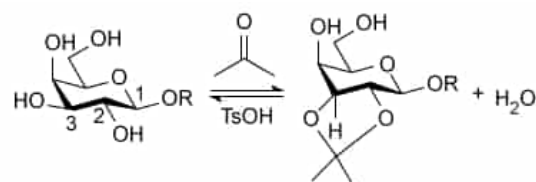


Figure 3 Acetonide protection of a galactose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

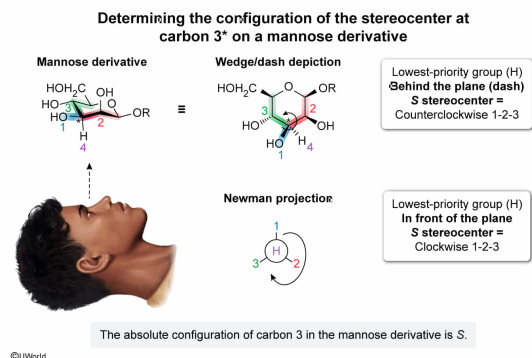
Based on the structure of the mannose derivative given in Figure 2, what is the absolute configuration of its carbon 3?

- ☐ A. L [10%]
- ☐ B. D [10%]
- ☐ C. R [35%]
- ☒ D. S [43%]

Correct

42% answered correctly

Explanation:



Sugars contain **chiral centers**, which are given an **R or S** designation based on Cahn-Ingold Prelog **priority rules**. Carbohydrate structures can be depicted as chair conformations, wedge/dash structures, or **Newman projections**.

The first step in determining the absolute configuration of carbon 3 on the mannose derivative given in Figure 2 is to assign priority (1–4) to each substituent:

- Carbon 3 is bonded to O, C, C, and H. O has the highest priority (1), and H has the lowest priority (4).
- The substituents on the two carbon atoms (carbons 2 and 4) are examined to determine priorities 2 and 3. Both carbons are bonded to O, C, and H, so the substituents of the next carbons in the chain (carbons 1 and 5) are used to break the tie. Carbon 5 is bonded to O, C, and H. Carbon 1 is bonded to O, O, and H (more bonds to O breaks the tie).
- Carbon 2 is priority 2, and carbon 4 is priority 3.

After assigning priority, the spatial arrangement of priorities 1, 2, and 3 can be determined using one of several approaches:

- The chair conformation can be **redrawn as a cyclohexane ring** with wedges and dashes to represent the stereochemistry of the substituents. The wedge/dash depiction for the mannose derivative carbon 3 has the **lowest-priority** group pointed **behind the plane** (dash) and priorities 1, 2, and 3 are in a **counterclockwise arrangement**.
- The configuration of carbon 3 can also be determined using a Newman projection. Looking at the chair conformation from the bottom down the H–carbon 3 bond, the **lowest priority group** (H) is the **front** atom. Priorities 1, 2, and 3 on the back C atom (carbon 3, circle in the Newman projection) are in a **clockwise arrangement**.

Therefore, carbon 3 has an **S configuration**.

(Choices A and B) The designations L and D are *relative* configurations of the highest number chiral carbon (in this case, carbon 5).

(Choice C) An *R* configuration is obtained if the priorities had the opposite arrangements but the lowest priority was in the same location in each depiction.

Educational objective:

Carbohydrate structures can be depicted using chair conformations, wedge/dash structures, or Newman projections. Determining a stereocenter's absolute configuration (ie, *R* or *S*) is done by assigning priority (1–4) to each substituent and examining the 1-2-3 arrangement of the substituents.

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Organic Chemistry

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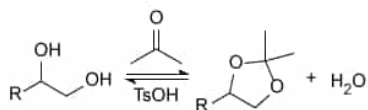


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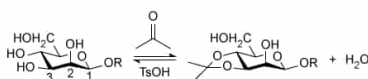


Figure 2 Acetonide protection of a mannose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

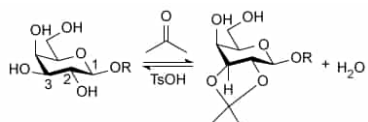


Figure 3 Acetonide protection of a galactose derivative (Note: Pyranose carbon atoms 1 to 3 are numbered; R = any alkyl group.)

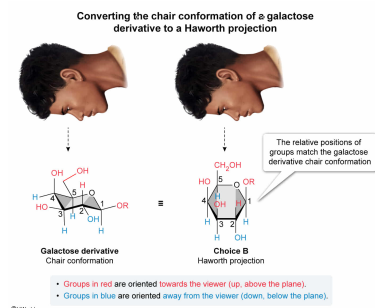
Which Haworth projection represents the galactose derivative shown in Figure 3?

- ☐ A. [5%]
- ✓ ☒ B. [70%]
- ☐ C. [11%]
- ☐ D. [12%]

Correct
70% answered correctly

Explanation:

explanation:



Carbohydrates can exist in an **open-chain or cyclic** form. The open chain form is often depicted in a **Fischer projection**, while the cyclic form (ie, **pyranose or furanose**) is often depicted in a chair conformation or Haworth projection. The cyclic form of sugars is formed by **cyclization** of the linear chain.

A **chair conformation** shows substituents in **axial or equatorial positions**. A **Haworth projection** is a flat depiction of the ring, where the bolded bonds are closest to the viewer, and substituents are either above the ring (pointing up) or below the ring (pointing down). Conversion between a chair conformation and Haworth projection requires placing the substituents on their corresponding carbon atom and **orienting the substituents in the correct up or down position**. Substituents that are **up** on the **chair conformation** are **also up** on the **Haworth projection** and vice versa for those in the down position.

If a viewer is looking from above down on the chair conformation of a galactose derivative in Figure 3, the nonhydrogen substituents have the following orientations:

- Carbons 1, 3, 4, and 5: toward the viewer (up)
- Carbon 2: away from the viewer (down)

The corresponding Haworth projection has the –OR and –OH substituents on carbons 1, 3, 4, and 5 pointed toward a viewer looking down on the ring from above (ie, oriented up, above the plane) and the –OH substituent on carbon 2 pointed away from the viewer (ie, oriented down, below the plane). **Choice B** is the Haworth projection that depicts the substituents in the correct orientation.

(Choice A) This Haworth projection **incorrectly positions** the –OH on carbon 2 *up*.

(Choice C) This Haworth projection **incorrectly positions** has the –OH on carbon 4 *down*.

(Choice D) This Haworth projection **incorrectly positions** both the –OH on carbon 2 *up* and the –OH on carbon 4 *down*. This is the correct depiction for the mannose derivative in Figure 2.

Educational objective:

Sugars can be depicted in linear (Fischer projection) and cyclic forms (chair conformation or Haworth projection). If a viewer is looking from above down on either cyclic depiction, the ring substituents are oriented either toward the viewer (up) or away from the viewer (down).

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Organic Chemistry

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System:
Functional Groups and Their Reactions